

$$R_1 = \frac{R_1 R_2}{R_1 + R_2 + R_3} \Leftrightarrow R_0 = R_1 + R_2 + \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

$$R_2 = \frac{R_1 R_2}{R_1 + R_2 + R_3}$$

串联电路: $\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{di}{dt} + \frac{1}{LC} i = 0$

$$\alpha = \frac{1}{2RC} \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

2. 电容 & 电感

$C = \frac{Q}{V}, W = \frac{1}{2} CV^2, i = C \frac{dV}{dt}$

串联同电感

$V = L \frac{di}{dt}, W = \frac{1}{2} Li^2$

$i \rightarrow + V$

Step response

串联: $\frac{d^2 V}{dt^2} + \frac{R}{L} \frac{dV}{dt} + \frac{V}{LC} = \frac{V_s}{LC}$

并联: $\frac{d^2 i}{dt^2} + \frac{1}{RC} \frac{di}{dt} + \frac{i}{LC} = \frac{I_s}{LC}$

电容两端电压
电感
相同

3. 一阶微分方程

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \left| \quad \frac{di}{dt} + \frac{1}{\tau} i = 0 \right.$$

$$\tau(x) = e^{\int P(x) dx} \quad i(t) = i(0) e^{-\frac{t}{\tau}}$$

$$y = \frac{1}{\tau(x)} \left[\int \tau(x) Q(x) dx + C \right]$$

source-free RC

$V(t) = V_0 e^{-t/RC} \quad (\tau = RC)$

电阻: $P(t) = \frac{V^2}{R} e^{-2t/RC}, W(t) = \frac{1}{2} CV_0^2 (1 - e^{-2t/RC})$

source-free RL

$i(t) = I_0 e^{-Rt/L} \quad (\tau = \frac{L}{R})$

电阻: $P(t) = I_0^2 R e^{-2Rt/L}, W(t) = \frac{1}{2} LI_0^2 (1 - e^{-2Rt/L})$

RC: $V(t) = V(\infty) + (V(0) - V(\infty)) e^{-t/\tau}$

RL: $i(t) = \frac{V(\infty)}{R} + (i(0) - \frac{V(\infty)}{R}) e^{-t/\tau}$

4. 二阶微分方程

source-free

串联: $\alpha = \frac{R}{2L}, \omega = \frac{1}{\sqrt{LC}}, S = -\alpha \pm \sqrt{\alpha^2 - \omega^2}$

S: natural frequency / characterist roots

1) overdamped: $\alpha > \omega_0$

$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$

2) critically damped: $\alpha = \omega_0$

$i(t) = e^{-\alpha t} (A_1 + A_2 t)$

3) underdamped: $\alpha < \omega_0$

$i(t) = e^{-\alpha t} (A_1 \cos \omega_d t + A_2 \sin \omega_d t)$

4) undamped: $\alpha = 0$

$\frac{d^2 i(t)}{dt^2} = \frac{V(0)}{L} = -\frac{(V_L(0) + i(0)R)}{L}$

正弦稳态分析

相位: the argument of sinusoid

$V_m \sin(\omega t + \phi) \rightarrow V_m \angle \phi$

impedance = $\begin{cases} j\omega L \\ \frac{1}{j\omega C} \end{cases} = R + jX$

admittance = $\frac{1}{Z}$

AC 功率分析

$P_{avg} = \frac{1}{2} V_m I_m \cos(\theta_V - \theta_I) \rightarrow \frac{P}{\sqrt{R^2 + X^2}}$

当 $Z_L = Z_{th}$ 时

$p = \frac{|V_{th}|^2}{8 R_{th}}$

apparent power $S = V_{rms} I_{rms}$

power factor $pf = \cos(\theta_V - \theta_I) \in [0, 1]$

complex power

$\vec{S} = V_{rms} \cdot \vec{I}_{rms}^* = \vec{I}_{rms}^2 \vec{Z} = \vec{I}_{rms}^2 (R + jX) = P + jQ$

Q reactive power, 单位是 VAR

$Q = 0$ resistive loads (unity pf)

$Q < 0$ capacitive loads (leading pf)

频率响应

$dB = 10 \lg \frac{P}{P_1}$

series resonance

$\omega_0 = \frac{1}{\sqrt{LC}}, H(\omega) = Z(\omega) = R$

bandwidth: $B = \omega_2 - \omega_1, (P_{max} = P_{min}) = \frac{P_m}{2}$

Quality factor

$Q = \frac{2\pi f_0 L}{R} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$

$B = \frac{R}{L} = \frac{\omega_0}{Q}$

parallel resonance

$Y(\omega) = H(\omega) = \frac{1}{R} + j\omega C + \frac{1}{j\omega L}$

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}$$

half-power frequencies:

$$\omega_{1,2} = \pm \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$B = \frac{1}{RC} = \frac{\omega_0}{Q}$$

$$Q = \frac{\omega_0}{B} = \omega_0 RC = \frac{R}{\omega_0 L}$$

8. ~~耦合~~ (coupled)

$$L = L_1 + L_2 + 2M$$

$$L = L_1 + L_2 - 2M$$

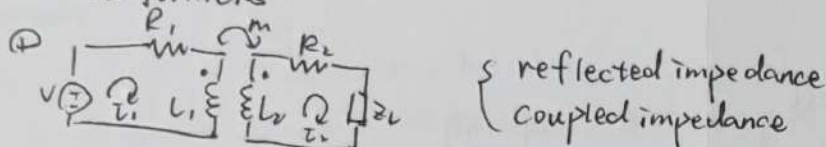
2. energy

$$W = \frac{1}{2} L_1 i_1^2 + \frac{1}{2} L_2 i_2^2 \pm M i_1 i_2$$

3. $M = k\sqrt{L_1 L_2}$ ($0 \leq k \leq 1$) $\rightarrow$ mutual inductance

$$k = \frac{\Phi_{12}}{\Phi_{11} + \Phi_{12}} \rightarrow \text{coupling coefficient}$$

Transformers



$$Z = \frac{V}{I} = R_1 + j\omega L_1 + \frac{\omega^2 M^2}{R_2 + j\omega L_2 + Z_L}$$



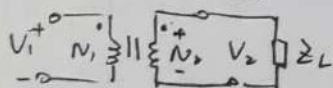
$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} L_A & L_C \\ L_C & L_B \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} j\omega(L_1 + L_2) & j\omega L_C \\ j\omega L_C & j\omega(L_1 + L_2) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$L_A = \frac{L_1 L_2 - M^2}{L_2 - M} \quad L_B = \frac{L_1 L_2 - M^2}{L_1 - M}$$

$$L_C = \frac{L_1 L_2 - M^2}{M}$$

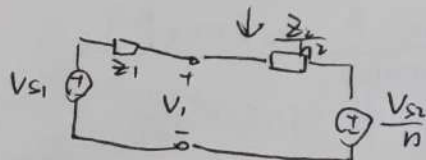
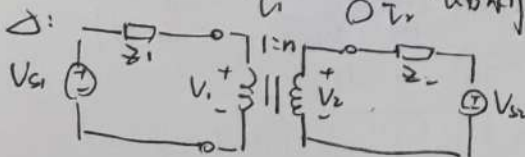
5. 理想变压器



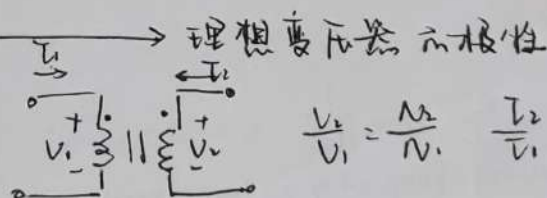
$$\frac{V_1}{V_2} = \frac{N_1}{N_2} + V_1 I_1 = V_2 I_2$$

$\therefore$ reflected im ~: $Z_{in} = \frac{Z_L}{n^2}$

$$\text{用 } Z_{in} = \frac{V_1}{I_1} = \frac{0 V_2}{0 I_2} \text{ 求分析}$$



$$\left(\frac{V_1}{V_{sr}} = \frac{1}{n} \right)$$



$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \quad \frac{I_2}{I_1} = -\frac{N_1}{N_2}$$

若 V_1, V_2 在同名端均为正/负, 则使用 $+n, -n$

若 I_1, I_2 都从同名端流入或流出, 则使用 $-n, +n$

